From Gaussians to Bayesian optimisation

From the bell curve to choosing the next experiment

Carlos Cotrini

ETH Zurich

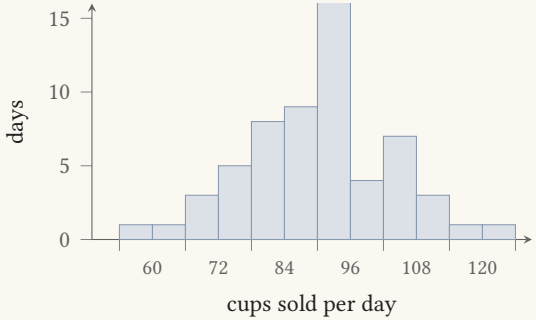
21 October 2026

1

Stage 1

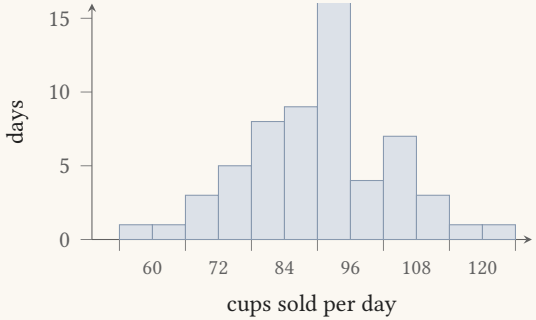
The 1D Gaussian

Daily sales, and a question



Real-world observations from 60 days.

Daily sales, and a question



Real-world observations from 60 days.

What is the chance that tomorrow we sell **between 90 and 120 cups**?

Prepare too little and we run out. Prepare too much and it melts.

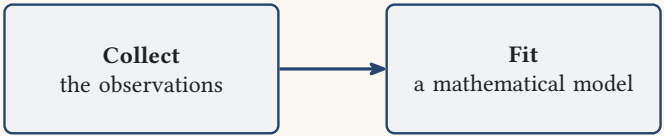
$$P(90 \leq X \leq 120) = ?$$

How data science answers it

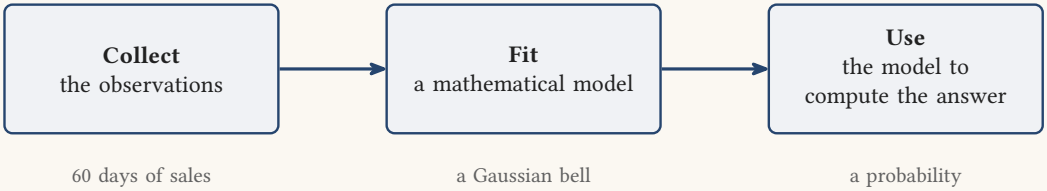
Collect
the observations

60 days of sales

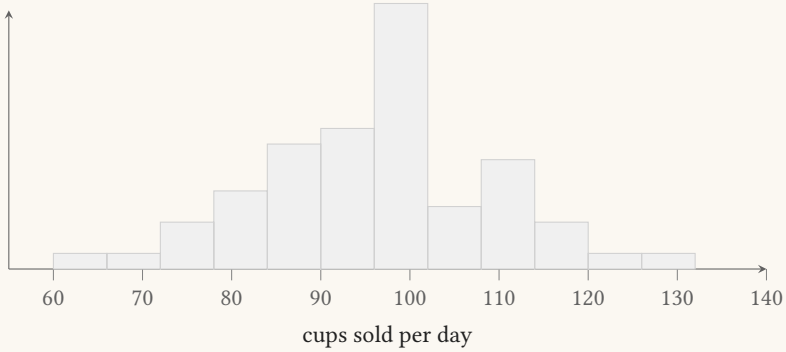
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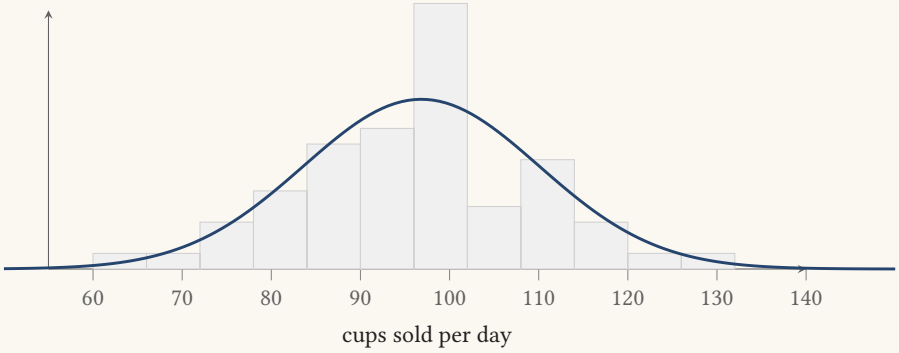
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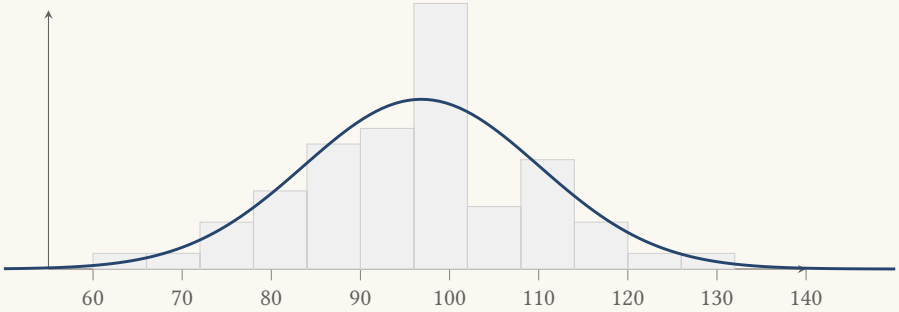
The model is a Gaussian



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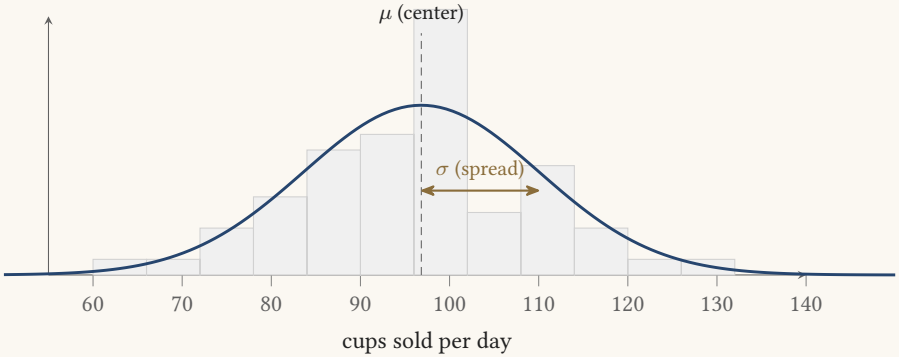


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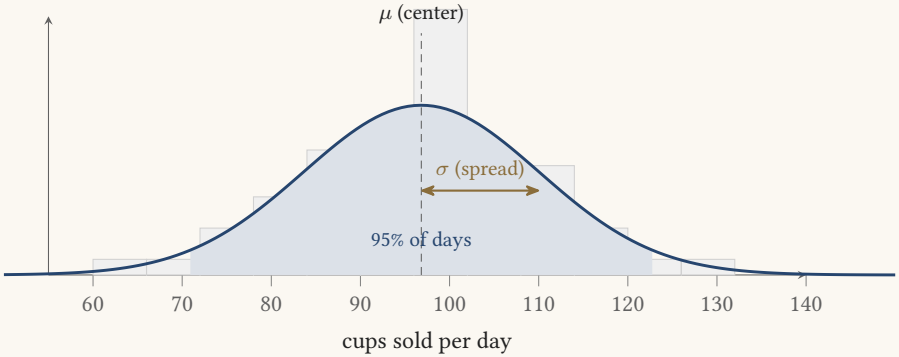
$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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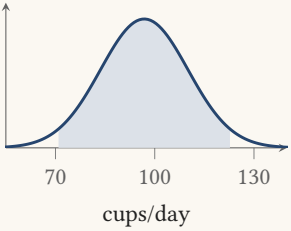
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Estimating the center and the spread

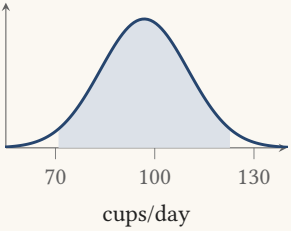


the fitted bell

Average of the data, which settles at the true mean:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i \xrightarrow[n\to\infty]{} \mathbb{E}[X] =: \mu$$

Estimating the center and the spread



the fitted bell

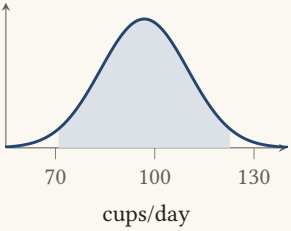
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$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} (x_i - \hat{\mu})^2 \xrightarrow[n\to\infty]{} \mathbb{E}\big[(X-\mu)^2\big] =: \sigma^2$$

Estimating the center and the spread



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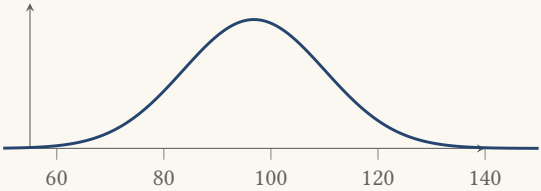
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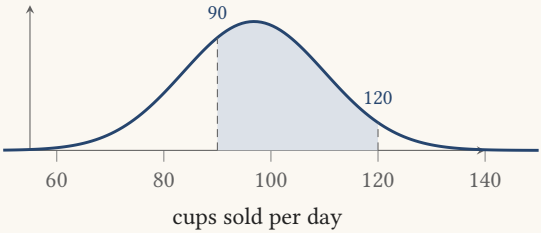
$$\hat{\mu} = 96.8 \qquad \hat{\sigma}^2 = 174.8$$

Answering the question



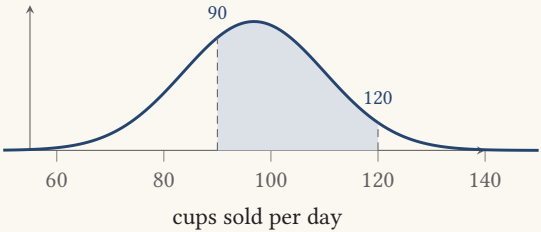
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Answering the question



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Answering the question



$$P(90 \le X \le 120) = \int_{90}^{120} g(x)\,dx$$

$$P = 0.66$$

about a 2-in-3 chance

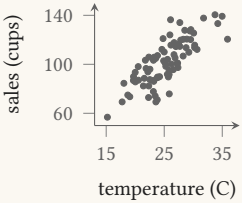
observations → a Gaussian model → a probability we can plan against

2

Stage 2

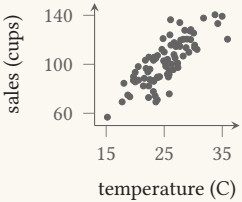
The 2D Gaussian

Three worlds, and a question

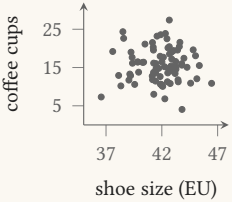


hotter → more sales

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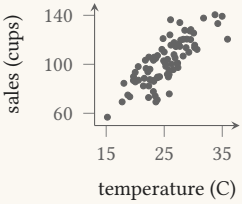


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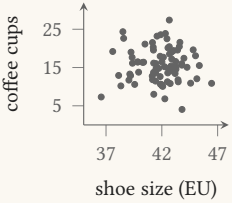


no relationship

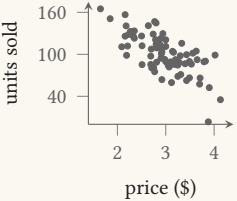
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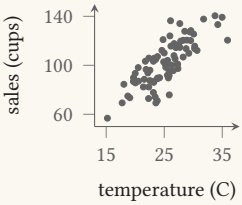


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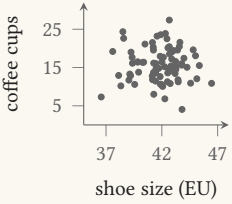


pricier → fewer sold

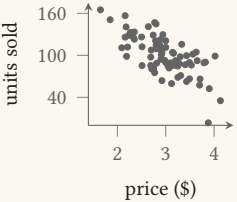
Three worlds, and a question



hotter → more sales



no relationship



pricier → fewer sold

On the first world, what is the chance a day is **warm and busy at once**?

temperature between 25 and 30 C, sales between 100 and 140 cups

$$P\big(\, T \in [25, 30] \;\text{ and }\; S \in [100, 140] \,\big) \;=\; ?$$

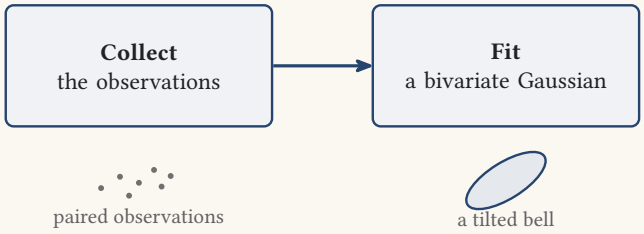
The same method, now in 2D

Collect
the observations



paired observations

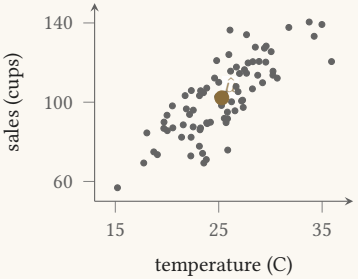
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The same method, now in 2D



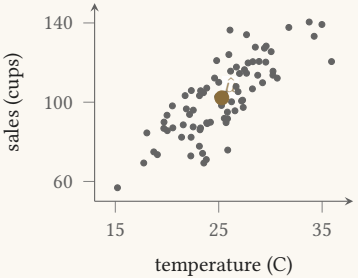
Estimating the pieces: center and per-axis spread



the mean, marked

Idea: run Stage 1 once on each axis.

Estimating the pieces: center and per-axis spread



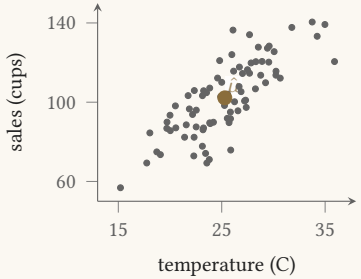
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Idea: run Stage 1 once on each axis.

The mean vector (average of each coordinate):

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\mu}_1 \\ \hat{\mu}_2 \end{pmatrix} = \begin{pmatrix} 25.3 \\ 102.2 \end{pmatrix}$$

Estimating the pieces: center and per-axis spread



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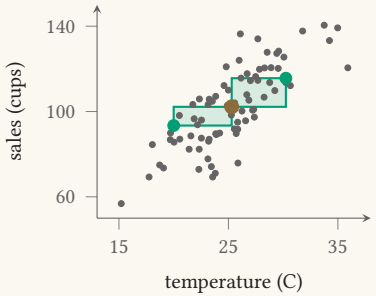
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The spread on each axis (variance):

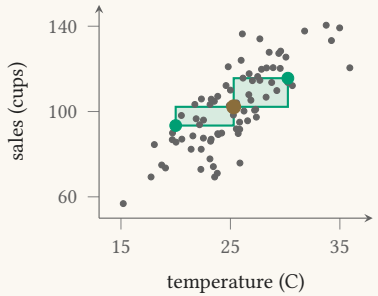
$\hat{\sigma}_1^2 = 16.0$ (temperature) $\quad\quad$ $\hat{\sigma}_2^2 = 348.2$ (sales)

Estimating the pieces: the correlation



two sample rectangles; signed area = a product

Estimating the pieces: the correlation



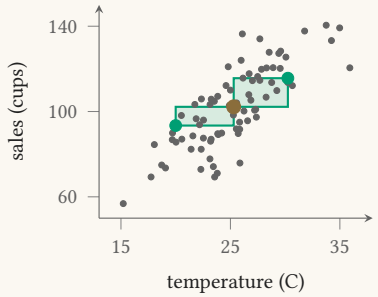
two sample rectangles; signed area = a product

Average of those signed areas:

$$\hat{\sigma}_{12} = \frac{1}{n}\sum_{i=1}^{n}(x_{1i} - \hat{\mu}_1)(x_{2i} - \hat{\mu}_2)$$

$$\xrightarrow[n\to\infty]{} \mathbb{E}\big[(X_1 - \mu_1)(X_2 - \mu_2)\big]$$

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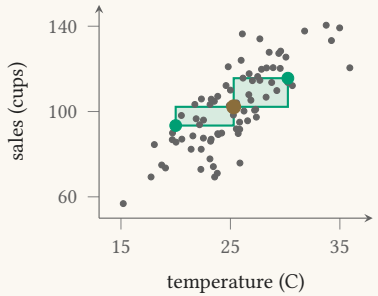
Its sign is the headline, across the three worlds:

positive $\hat{\sigma}_{12} = +58.8$, corr $+0.79$

neutral $\hat{\sigma}_{12} \approx 0$, corr 0.00

negative $\hat{\sigma}_{12} = -9.7$, corr -0.70

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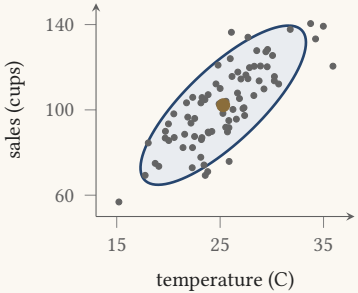
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The scale-free reading is the correlation, always in $[-1, 1]$.

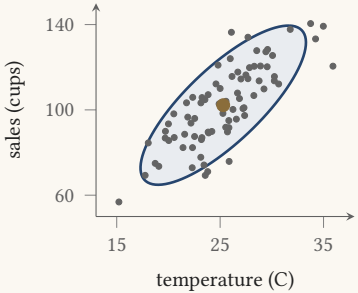
The bivariate Gaussian, and its covariance matrix



the fitted bell, tilted by + correlation

with the point $\mathbf{x} = (x_1, x_2)^\top$ and center $\boldsymbol{\mu} = (\mu_1, \mu_2)^\top$:

The bivariate Gaussian, and its covariance matrix

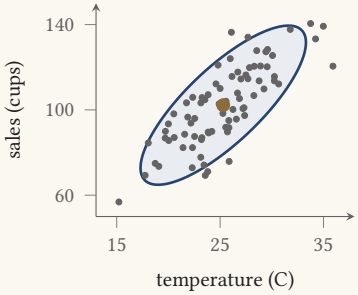


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$$g(\mathbf{x}) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big)$$

The bivariate Gaussian, and its covariance matrix



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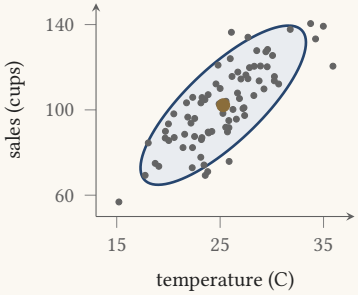
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Σ is the covariance matrix: spreads on the diagonal, covariance off it.

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}$$

The bivariate Gaussian, and its covariance matrix



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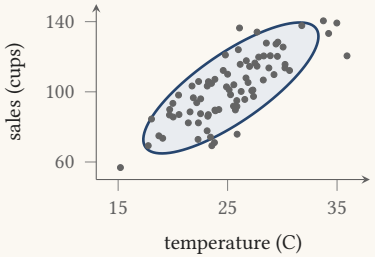
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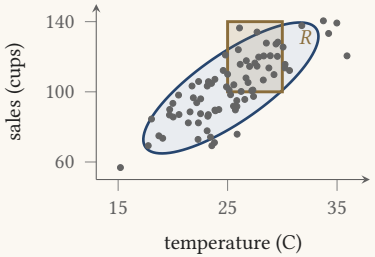
The answer: a double integral over the region



the region $R = [25, 30] \times [100, 140]$

Chance the day lands inside the rectangle R:

The answer: a double integral over the region

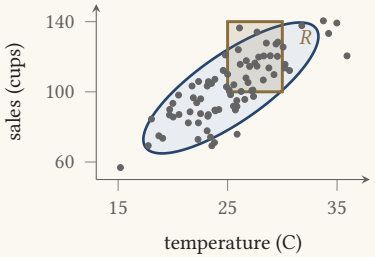


the region $R = [25, 30] \times [100, 140]$

Chance the day lands inside the rectangle R:

$$P\big((X_1, X_2) \in R\big) = \iint_R g(x_1, x_2)\, dx_1\, dx_2$$

The answer: a double integral over the region



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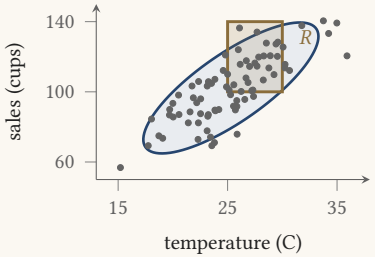
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$$P\big((X_1, X_2) \in R\big) = \iint_R g(x_1, x_2)\, dx_1\, dx_2$$

$$P = 0.31$$

about a 1-in-3 chance

The answer: a double integral over the region



the region $R = [25, 30] \times [100, 140]$

Chance the day lands inside the rectangle R:

$$P\big((X_1, X_2) \in R\big) = \iint_R g(x_1, x_2)\, dx_1\, dx_2$$

$$P = 0.31$$

about a 1-in-3 chance

two questions at once → a bivariate Gaussian (μ, Σ from data) → a probability over a region

3

Stage 3

The multivariate Gaussian

Three flavors over 40 days, and a question

day	vanilla	chocolate	strawberry
1	55	41	35
2	64	44	36
3	41	29	30
4	67	36	39
5	59	41	35
6	73	53	37
⋮	⋮	⋮	⋮

Each number is cups sold that day. 40 days in total.

Each row is one day; each number is cups sold. Hot days lift all three flavors together.

Three flavors over 40 days, and a question

day	vanilla	chocolate	strawberry
1	55	41	35
2	64	44	36
3	41	29	30
4	67	36	39
5	59	41	35
6	73	53	37
⋮	⋮	⋮	⋮

Each number is cups sold that day. 40 days in total.

Each row is one day; each number is cups sold. Hot days lift all three flavors together.

We plan a target range for each flavor. Call a day **balanced** when all three land inside their target at once, so nothing runs out and nothing is left over:

vanilla 40 to 60, chocolate 30 to 50, strawberry 20 to 40 cups

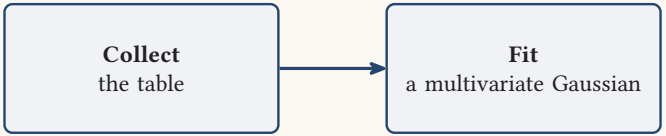
$$P(\text{a balanced day}) = ?$$

The same method, in three dimensions

Collect
the table

40 days, 3 flavors

The same method, in three dimensions



40 days, 3 flavors

a mean vector and
a covariance matrix

The same method, in three dimensions



A mean vector and a covariance matrix

The mean vector, one average per flavor (per column of the table):

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} 53.4, & 41.5, & 31.4 \end{pmatrix}$$ (vanilla, chocolate, strawberry), in cups

A mean vector and a covariance matrix

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The covariance matrix $\hat{\Sigma}$ (spreads on the diagonal):

	vanilla	chocolate	strawberry
vanilla	91.6	66.7	51.8
chocolate	66.7	108.0	49.6
strawberry	51.8	49.6	51.1

all nine entries positive: the flavors move together.

A mean vector and a covariance matrix

The mean vector, one average per flavor (per column of the table):

$$\hat{\boldsymbol{\mu}} = \big(\,53.4,\;\; 41.5,\;\; 31.4\,\big)$$ (vanilla, chocolate, strawberry), in cups

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vanilla	91.6	66.7	51.8
chocolate	66.7	108.0	49.6
strawberry	51.8	49.6	51.1

all nine entries positive: the flavors move together.

The general Gaussian density, for a point $\mathbf{x}$ in d dimensions:

$$g(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}\sqrt{|\Sigma|}}\;\exp\!\Big(-\tfrac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\Big)$$

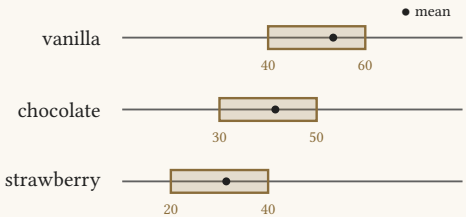
Here $d = 3$. (Stage 1 was $d = 1$, Stage 2 was $d = 2$; only the size of $\boldsymbol{\mu}$ and Σ changes.)

The answer: a multi-dimensional integral



Chance every flavor lands in its target range at once (a balanced day):

The answer: a multi-dimensional integral



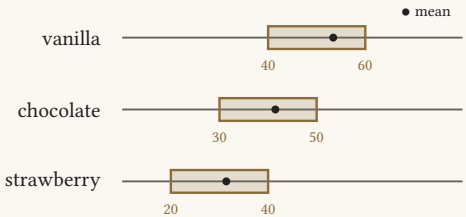
each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

$$P(\mathbf{X} \in R) = \iiint_R g(x_1, x_2, x_3)\, dx_1\, dx_2\, dx_3$$

the balanced region $R = [40, 60] \times [30, 50] \times [20, 40]$ cups

The answer: a multi-dimensional integral



each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

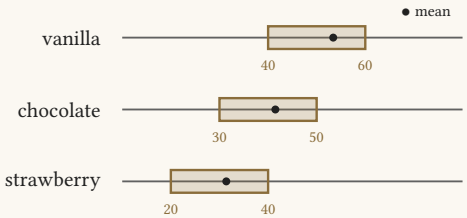
$$P(\mathbf{X} \in R) = \iiint_R g(x_1, x_2, x_3)\, dx_1\, dx_2\, dx_3$$

the balanced region $R = [40, 60] \times [30, 50] \times [20, 40]$ cups

$$P = 0.48$$

about half our days are balanced; the rest miss on a flavor

The answer: a multi-dimensional integral



each bar is one flavor; the shaded band is its target range.

Chance every flavor lands in its target range at once (a balanced day):

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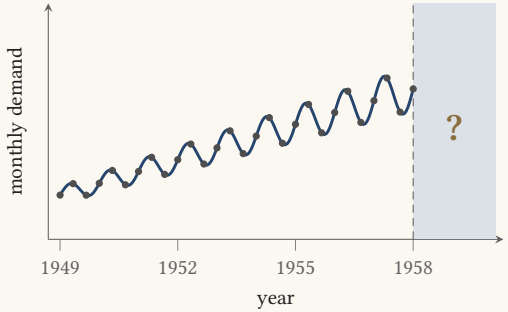
many quantities at once → a multivariate Gaussian ($\boldsymbol{\mu}$ vector, Σ matrix) → a probability over a region

4

Stage 4

The Gaussian process

Demand through 1958, and a question

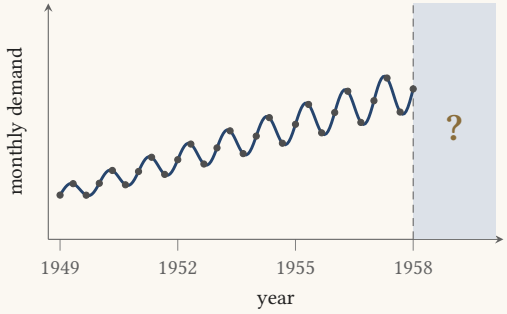


Monthly demand through 1958; the next two years are blank.

We track a monthly **demand** series through 1958.

Real airline-passenger numbers stand in for a shop's revenue series.

Demand through 1958, and a question



Monthly demand through 1958; the next two years are blank.

We track a monthly **demand** series through 1958.

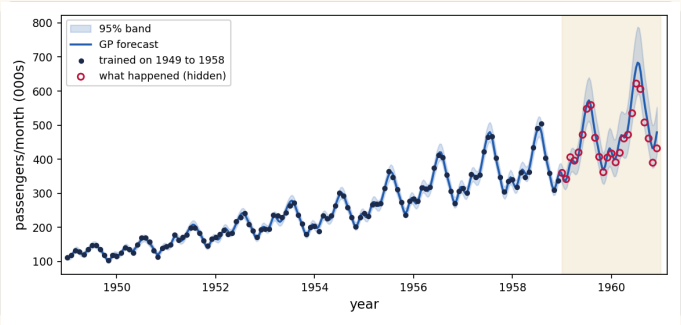
Real airline-passenger numbers stand in for a shop's revenue series.

What will demand be over the **next two years**?

And how **sure** can we be?

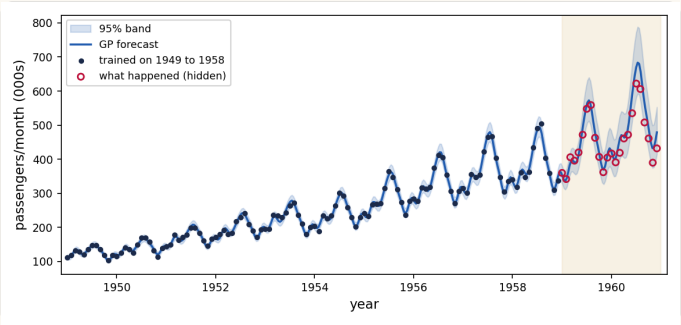
We want an honest range, not just a single number.

Fit a Gaussian process: the forecast



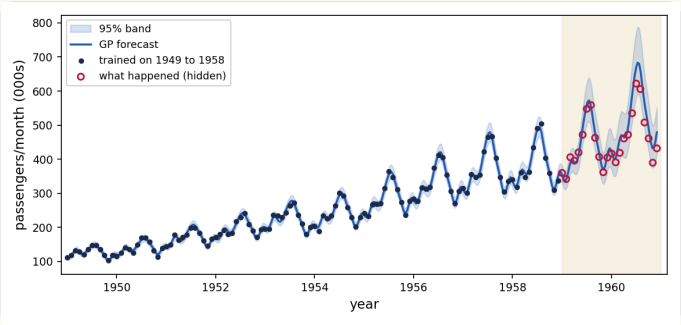
Trained only on the **navy dots**: monthly demand, 1949 to 1958.

Fit a Gaussian process: the forecast



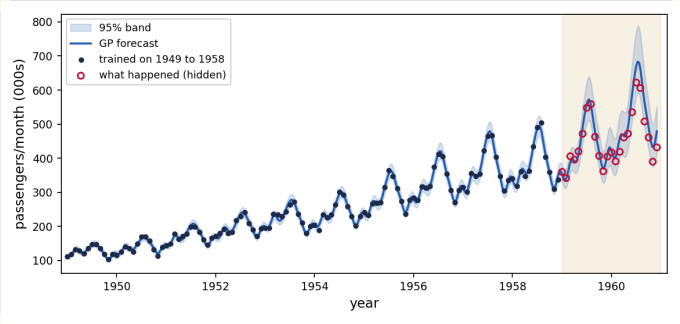
The **blue line and band** are the two-year forecast that follows.

Fit a Gaussian process: the forecast



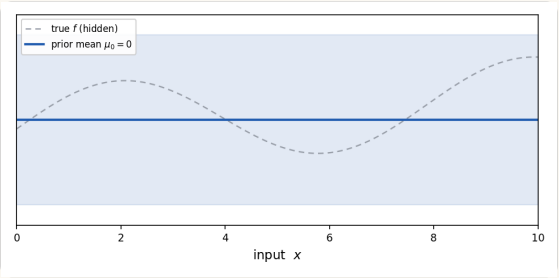
The **red circles** are the hidden truth, every one **inside the band** (≈4.6% error).

Fit a Gaussian process: the forecast



A forecast that also says how sure it is.

So what is a Gaussian process?



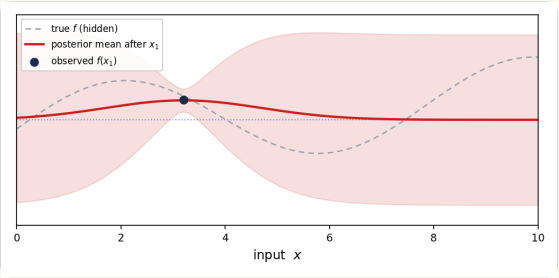
Before any data: the mean is a **flat guess** μ_0 and the band is **wide**. We know almost nothing.

A Gaussian process is just **two** things:

1. A mean function $m(x) = \mathbb{E}[f(x)]$

the best guess at every input (flat before data).

So what is a Gaussian process?



Pick x_1, observe $f(x_1)$: the mean **bends** to it, and the band **pinches** at the point.

A Gaussian process is just **two** things:

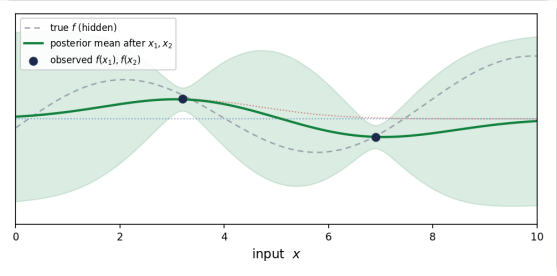
1. A mean function $m(x) = \mathbb{E}[f(x)]$

the best guess at every input (flat before data).

2. A kernel $k(x, x')$

how similar two inputs are; nearby inputs → correlated, so the mean bends and the band pinches at the data.

So what is a Gaussian process?



Pick x_2, observe $f(x_2)$: the mean **threads both**, and the band tightens. More data, less doubt.

A Gaussian process is just **two** things:

1. A mean function $m(x) = \mathbb{E}[f(x)]$

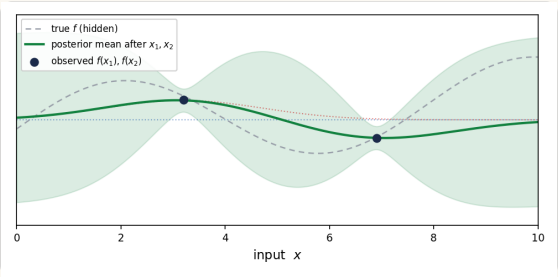
the best guess at every input (flat before data).

2. A kernel $k(x, x')$

how similar two inputs are; nearby inputs → correlated, so the mean bends and the band pinches at the data.

The **band** is the leftover uncertainty: small at the data, wide away.

So what is a Gaussian process?



Pick x_2, observe $f(x_2)$: the mean **threads both**, and the band tightens. More data, less doubt.

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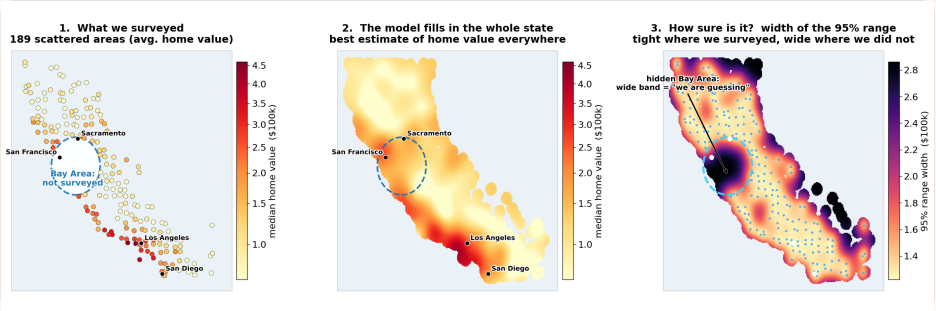
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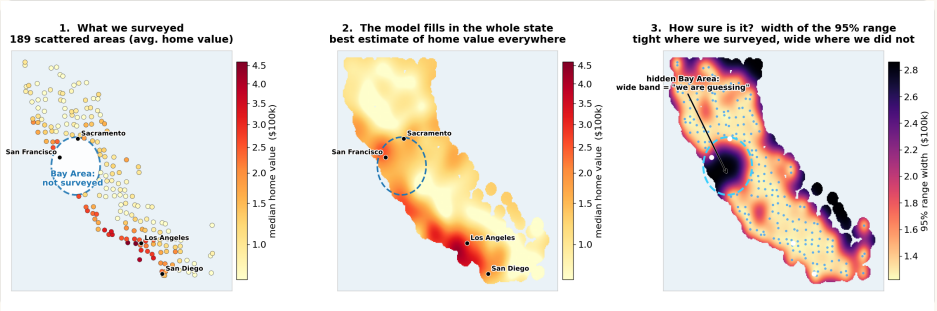
We infer m and k by Bayesian inference, which we do not cover here; the library does it.

The same idea, over a map



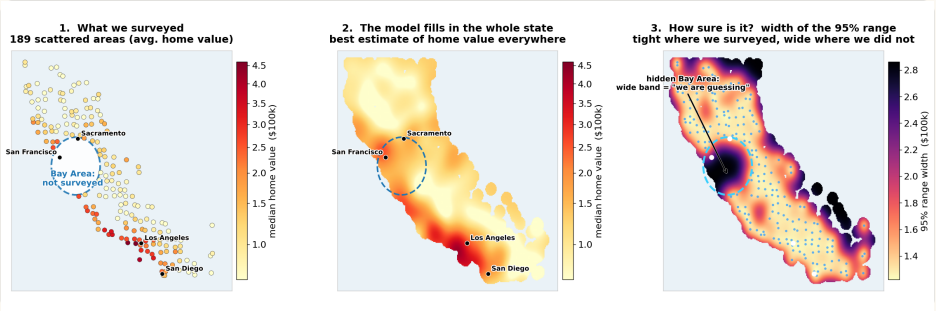
Now the input is a **place**; the kernel compares two locations.

The same idea, over a map



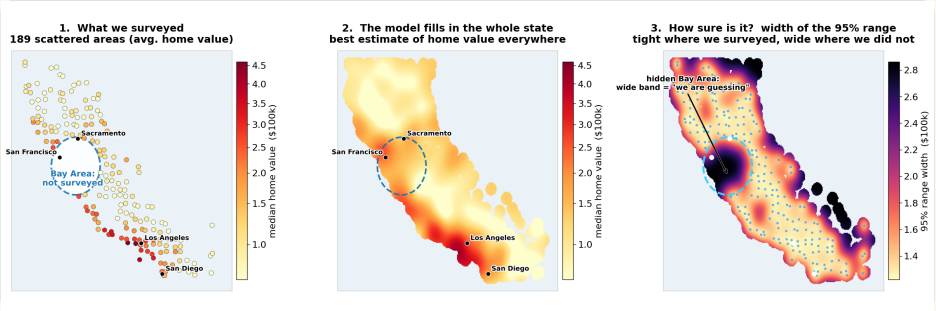
From a scattered survey, one Gaussian process **fills in the whole state**.

The same idea, over a map



Panel 3 **flags the guessing**: ≈1.8x wider over the Bay Area, still **88% covered**.

The same idea, over a map



The same Gaussian process, a different input space.

One idea, many input spaces

A time
a demand forecast

input = a time

One idea, many input spaces

A time
a demand forecast

input = a time

A place
a filled-in map

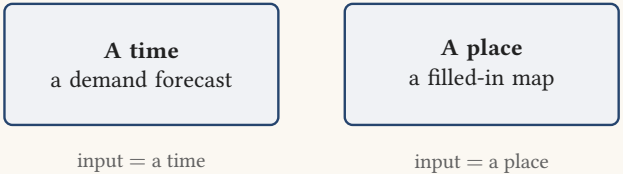
input = a place

One idea, many input spaces



In code it is a few lines: choose a kernel, fit, then predict with a band.

One idea, many input spaces



In code it is a few lines: choose a kernel, fit, then predict with a band.

a mean function + a kernel → predict anywhere, with honest error bars

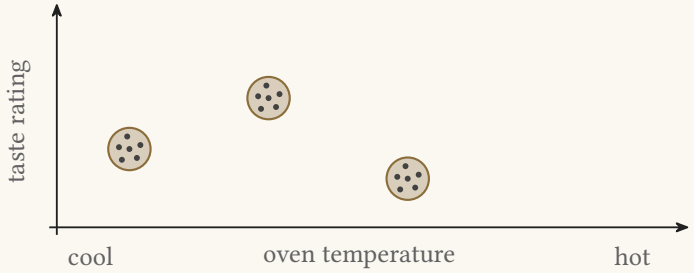
5

Stage 5

Bayesian optimisation

Slow bakes, few tries

Finding the best oven temperature for cookies.

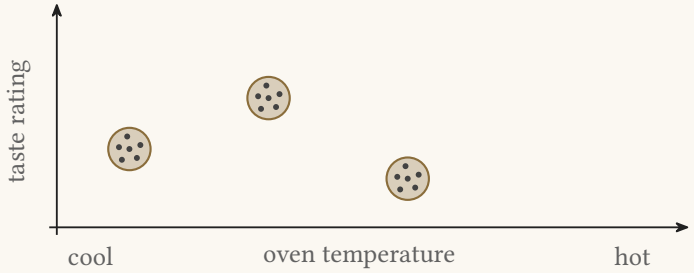


1. Each cookie: one bake, one rating.

Kochanski et al. 2017; toy: ours

Slow bakes, few tries

Finding the best oven temperature for cookies.



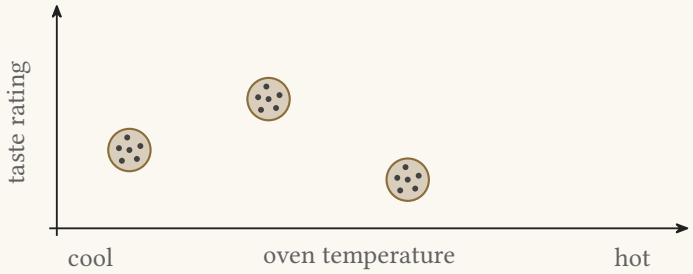
1. Each cookie: one bake, one rating.

2. Between them: unknown.

Kochanski et al. 2017; toy: ours

Slow bakes, few tries

Finding the best oven temperature for cookies.



1. Each cookie: one bake, one rating.

2. Between them: unknown.

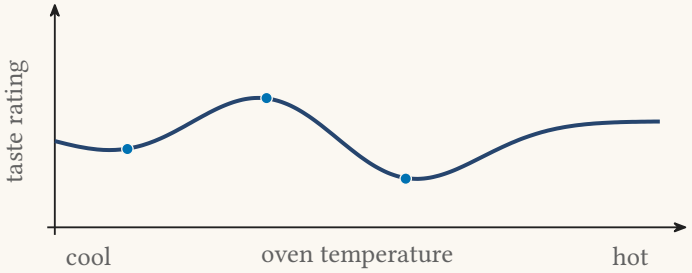
3. A sweep fixes every temperature in advance.

Let each taste guide the next bake.

Kochanski et al. 2017; toy: ours

A cheap model learns from bakes

Ask it about untried temperatures.

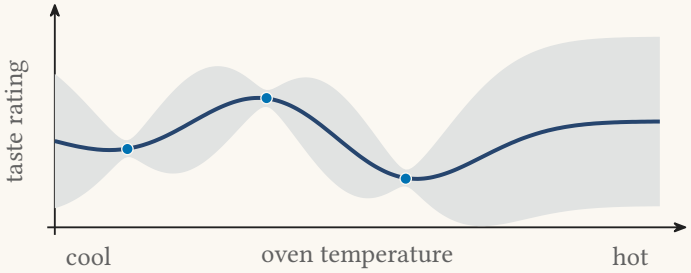


1. Line: the prediction.

Frazier 2018; Krause and Hübotter; toy: ours

A cheap model learns from bakes

Ask it about untried temperatures.



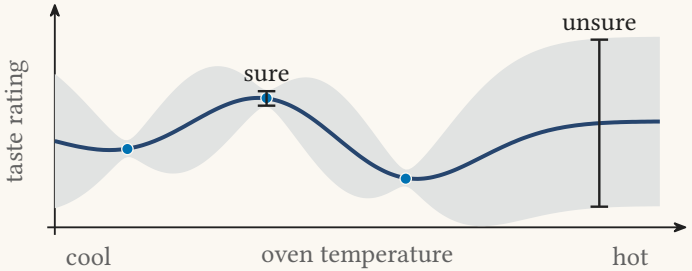
1. Line: the prediction.

2. Band: how unsure.

Frazier 2018; Krause and Hübotter; toy: ours

A cheap model learns from bakes

Ask it about untried temperatures.



1. Line: the prediction.

2. Band: how unsure.

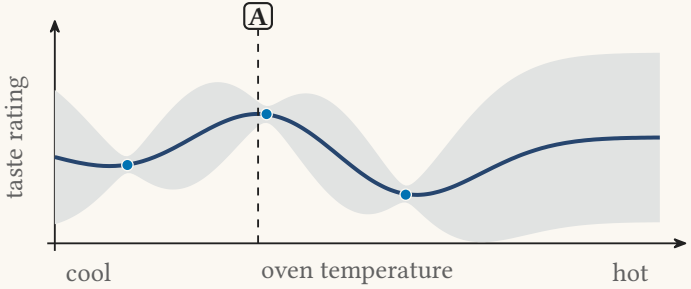
Surrogate model: a cheap stand-in, often a Gaussian process.

It predicts and says how sure.

Frazier 2018; Krause and Hübotter; toy: ours

Try the promising, or the unknown

Two reasons to try a temperature.

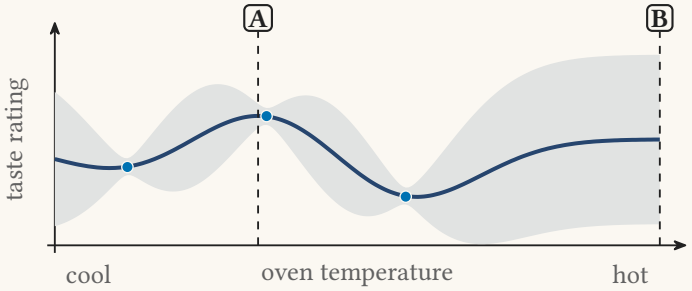


1. A exploits: the best guess.

Krause and Hübotter; Brochu et al. 2010; toy: ours

Try the promising, or the unknown

Two reasons to try a temperature.



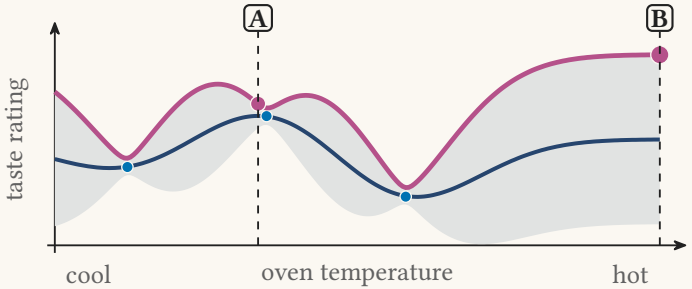
1. A exploits: the best guess.

2. B explores: the widest band.

Krause and Hübotter; Brochu et al. 2010; toy: ours

Try the promising, or the unknown

Two reasons to try a temperature.



Acquisition function: a score per setting.

1. A exploits: the best guess.

2. B explores: the widest band.

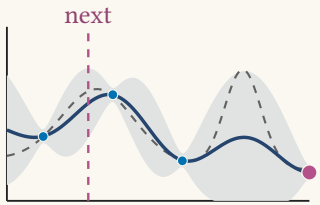
3. Score = top of the band.

Bake at the highest score.

Krause and Hübotter; Brochu et al. 2010; toy: ours

Fit, pick, bake, repeat

One search, three rounds. Dashed: the true curve, unseen.

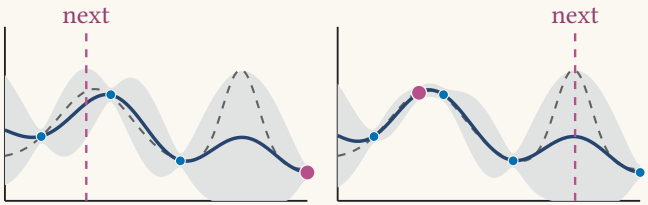


1. Hottest oven: burnt.

Brochu et al. 2010; Shahriari et al. 2016; toy: ours

Fit, pick, bake, repeat

One search, three rounds. Dashed: the true curve, unseen.

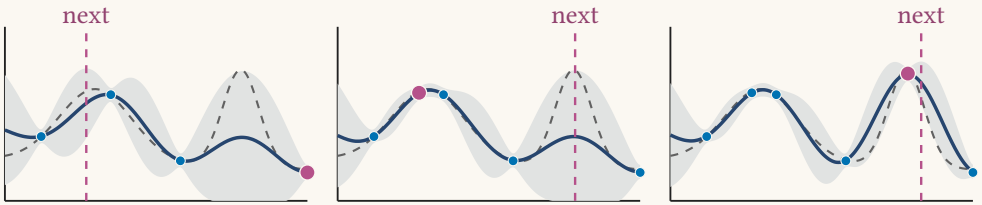


1. Hottest oven: burnt. **2.** Beside the best: same.

Brochu et al. 2010; Shahriari et al. 2016; toy: ours

Fit, pick, bake, repeat

One search, three rounds. Dashed: the true curve, unseen.



1. Hottest oven: burnt. **2.** Beside the best: same. **3.** In between: best yet.

This loop is Bayesian optimisation.

Brochu et al. 2010; Shahriari et al. 2016; toy: ours